Liyan Wu

Availability: Full-time industry or research internships · Nov–Dec 2026 start · Six months.

diana@hnu.edu.cn · +86 19174911723 · Changsha, China
GitHub: S1yang · OpenReview

Research Interests

Efficient and reliable LLM/VLM inference; speculative decoding and inference systems; multimodal hallucination mitigation; agent efficiency.

Education

Hunan University — Changsha, China
Master's Program in Artificial Intelligence · September 2024–June 2027 (expected)
College of Computer Science and Electronic Engineering · Advisor: Fan Wu
GPA: **3.50/4.00** · Overall average: **87.40/100**

Nanjing University of Science and Technology — Nanjing, China
Bachelor of Engineering in Software Engineering · September 2019–June 2023
School of Computer Science and Engineering · GPA: **3.18/4.00**

Publications & Manuscripts

Accepted Papers

[1] Mitigating Object Hallucinations in Vision-Language Models via Head-Level Attribution and Counteractive Pruning
Liyan Wu, Xin Luo, Fan Wu, Yingen Liu, Xiaobin Wu.
Findings of EMNLP 2026 — Accepted. **Co-first author, listed first.**

[2] MCTQ: Modality-Coordinated Transformation Quantization for Multimodal Large Language Models
Yingen Liu, Lanfeng Wu, Fan Wu, **Liyan Wu**, Ruihui Li, Zhuo Tang, Eng Siong Chng, Kenli Li.
EMNLP 2026 Main Conference — Accepted.

Manuscripts

[3] Beyond Local Evidence: Relay-Aware Verification for Block-Parallel Speculative Decoding
Liyan Wu — First author.
Under review at NAACL 2027.

[4] Relaxing the Object Hallucination–Utility Trade-off in Image Captioning via Dynamic Pathway Rebalancing
Liyan Wu — First author.
Manuscript; planned submission to CVPR 2027.

[5] Not All Mismatches Are Equal: Context-Aware Verification for Multimodal Speculative Decoding
Liyan Wu — Third author.
Manuscript; planned submission to CVPR 2027.

Research Experience

Hunan University · Advisor: Fan Wu

Object Hallucination Mitigation in Vision–Language Models

April 2025–May 2026

- Led HACP and DPR research from problem formulation and method design through implementation, evaluation, manuscript writing, and visualization, focusing on reducing object hallucinations while preserving descriptive coverage and utility.
- **HACP:** Developed head-level attribution and adaptive intervention without retraining. On a 1,000-image COCO subset, reduced CHAIR-i from **18.0% to 8.6%** and improved object-description F1 from **73.0 to 77.0**.
- **DPR:** Developed input-adaptive attention pathway rebalancing with an explicit generation budget. With at most three complete caption attempts on LLaVA-1.5-7B, reduced CHAIR-i from **12.77% to 6.46%** and improved F1 from **73.79 to 76.57**.

LLM & Multimodal Inference Acceleration

December 2025–Present · RAV research since May 2026

- **RAV — Method design:** Designed Relay-Aware Verification for block-parallel speculative decoding, combining target-model support for a draft token with confidence in the subsequent prediction to accept useful candidates without additional target-model forward passes.
- **RAV — Systems implementation:** Implemented the Transformers reference, integrated the acceptance rule into vLLM's Triton verification kernel, and incorporated DFlash and DSpark generation workflows. Checked GPU-kernel outputs against the reference implementation.
- **RAV — Evaluation:** Across 50 settings covering 4B–27B target models and five benchmarks on a single H100 80GB with BF16 greedy, single-request decoding, increased mean accepted length by **23.9%** and mean throughput by **32.3%** over the corresponding strict-verification baselines. Across model configurations, task-averaged quality metrics decreased by **0.9%–3.3%** relative to those baselines.
- **MCTQ:** Contributed ablation studies and efficiency analysis across models and quantization settings, examining how modality-gradient coordination affects low-bit accuracy and inference cost. The shared transformation introduces no additional inference overhead.
- **CAVE:** Contributed multimodal evidence alignment, mismatch-risk design, method figures, and LVSpec baseline evaluation. The collaborative study increased mean accepted tokens per verification step by **17.6%** over strict ViSpec across four models and five benchmarks, with up to **2.24×** speedup over autoregressive decoding while preserving task performance.

Agent Evaluation, Execution Profiling & Adaptive Inference

May 2026–Present

- Co-designed execution tracing and cost profiling at token, model-call, and episode levels; extended the τ^2-bench adapter and implemented an AppWorld adapter in an Inspect AI–based evaluation framework.
- Developed analysis components, bounded retries, error logs, execution tests, and run-archive validation to support reproducible benchmarking and failure diagnosis. An AppWorld integration run completed all **57 development samples** and **2,005 tool calls** with **zero Inspect transport failures**.
- Took primary responsibility for research questions, technical design, proposal writing, and figures for a **2027 Hunan Provincial Natural Science Foundation General Program application**. The proposal investigates quality-constrained inference strategy selection and resource scheduling for multimodal and agent workloads, including cache prefetching, tool warmup, and state reuse.

Technical Skills

- **Programming & ML:** Python, PyTorch, Transformers; attention interventions, speculative verification, and GPU-kernel correctness checks.
- **Inference & Evaluation:** vLLM, SGLang, Triton verification-kernel integration, Inspect AI, τ^2-bench, AppWorld; throughput–quality evaluation and execution profiling.
- **Experimentation & Engineering:** pandas, Jupyter, Matplotlib; Ubuntu/Linux, Shell, Git, Docker, Docker Compose, pytest; experiment automation, ablations, and result visualization.
- **Languages:** Chinese (native); English (CET-6), with research manuscript and technical documentation writing experience.